

Lecture 4: SVD, matrix norms

SVD = Singular Value Decomposition

Theorem: Every matrix $A \in \mathbb{R}^{m \times n}$ can be decomposed as $A = U \Sigma V^T$,

where $U \in \mathbb{R}^{m \times m}$ orthogonal, $V \in \mathbb{R}^{n \times n}$ orthogonal,

$\Sigma \in \mathbb{R}^{m \times n}$ diagonal matrix = $\text{diag}(\sigma_1, \dots, \sigma_{\min\{n, m\}})$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min\{n, m\}} \geq 0.$$

Remark:

Singular values $\sigma_1, \dots, \sigma_{\min\{n, m\}}$ are unique, while U and V , in general, are not.

A Square $\boxed{A} = \boxed{U} \boxed{\begin{smallmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{smallmatrix}} \boxed{V^T}$

A $\begin{smallmatrix} \boxed{} \\ (m > n) \end{smallmatrix} \quad \boxed{A} = \boxed{U} \boxed{\begin{smallmatrix} \sigma_1 \\ \vdots \\ \sigma_n \end{smallmatrix}} \boxed{V^T}$

A $\begin{smallmatrix} \boxed{} \\ (m < n) \end{smallmatrix} \quad \boxed{A} = \boxed{U} \boxed{\begin{smallmatrix} \sigma_1 & \dots & \sigma_m \end{smallmatrix}} \boxed{V^T}$

Def: $\sigma_1, \sigma_2, \dots, \sigma_{\min\{n, m\}}$ are the SINGULAR VALUES of A .

Columns of U ($u_1, u_2, \dots, u_m$) are the LEFT SINGULAR VECTORS of A

Columns of V ($v_1, v_2, \dots, v_n$) are the RIGHT SINGULAR VECTORS of A .

Remark: if $A \in \mathbb{R}^{n \times n}$ $A = U \Sigma V^T \Rightarrow A^T A = (U \Sigma V^T)^T (U \Sigma V^T)$

$$= V \underbrace{\Sigma^T U^T U}_{I} \Sigma V^T = V \underbrace{\Sigma^2}_{\text{spectral decomposition of } A^T A} V^T$$

Symmetric positive semi-definite

$\sigma_1, \sigma_2, \dots, \sigma_n$ are the singular values of A

$\Leftrightarrow \sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$ are the eigenvalues of $A^T A$.

Proof in the case of square matrix $A \in \mathbb{R}^{n \times n}$:

Let $A^T A = Q D Q^T$ be a spectral decomposition of $A^T A$

Let's take $V = Q \quad \Sigma = \sqrt{D} = \begin{bmatrix} \sqrt{d_1} & & \\ & \sqrt{d_2} & \\ & & \ddots \end{bmatrix}$

(if we want $A = U \Sigma V^T \Leftrightarrow A V = U \Sigma \cancel{V^T V} \Leftrightarrow U = A V \Sigma^{-1}$)

Let's take $U := A Q D^{-1/2}$.

Claim: $U \Sigma V^T$ is an SVD of A :

- $U \Sigma V^T = A Q \cancel{D^{-1/2} D^{1/2}} Q^T = A Q Q^T = A$
- Σ is diagonal with nonincreasing nonnegative diagonal entries
- V is orthogonal
- U is orthogonal: $U U^T = A Q D^{-1/2} D^{-1/2} Q^T A^T = A A^T$ (?)
 $U^T U = D^{-1/2} Q^T A^T A Q D^{-1/2} = I$

$$= D^{-1/2} \underbrace{Q^T Q}_I D \underbrace{Q Q^T}_I D^{-1/2} = D^{-1/2} D D^{-1/2} = I \quad \square$$

Def. (equivalent to the one seen in Lecture 1):

$\text{rank}(A) = \# \text{ of nonzero singular values}$

$$A = \begin{bmatrix} | & & | \\ u_1 & u_2 & \dots & u_m \\ | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_r & \\ & & 0 \end{bmatrix} \begin{bmatrix} | & & | \\ v_1^T & & v_n^T \\ | & & | \end{bmatrix} \quad \text{with } r \text{ nonzero } \sigma_i$$

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T \quad \rightarrow \text{sum of } r \text{ rank-1 terms.}$$

$\boxed{\quad} =$

Remark: to write A we only need the first $r = \text{rank}(A)$ singular values and left & right singular vectors!

$$A = \begin{bmatrix} | & & | \\ u_1 & u_2 & \dots & u_r & u_{r+1} & \dots & u_m \\ | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_r & \\ & & 0 \end{bmatrix} \begin{bmatrix} | & & | \\ v_1^T & & v_n^T \\ | & & | \end{bmatrix}$$

The SVD gives ("for free") information on the 4 fundamental subspaces of A :

- $\text{Im}(A) = \text{span}\{u_1, \dots, u_r\}$
- $\text{Ker}(A) = \text{span}\{u_{r+1}, \dots, u_m\}$
- $\text{Im}(A^T) = \text{span}\{v_1, \dots, v_r\}$
- $\text{Ker}(A^T) = \text{span}\{v_{r+1}, \dots, v_n\}$

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$$

$$A \cdot v_{r+1} = \sigma_1 u_1 \underbrace{v_1^T v_{r+1}}_0 + \dots + \sigma_r u_r \underbrace{v_r^T v_{r+1}}_0 = 0$$

Def: The thin/economy SVD of a rectangular $A \in \mathbb{R}^{m \times n}$ ($m > n$)

$$\text{is } A = \begin{bmatrix} | & & | \\ u_1 & & u_n \\ | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_n & \\ & & 0 \end{bmatrix} \begin{bmatrix} | & & | \\ v_1^T & & v_n^T \\ | & & | \end{bmatrix}$$

↳ orthonormal columns

Questions:

(a) Given $A = U \Sigma V^T$, can you find an SVD of A^T ? $A^T = (U \Sigma V^T)^T = \overset{\text{orth.}}{\downarrow} V \overset{\text{orth.}}{\downarrow} \overset{\text{diag.}}{\uparrow} \Sigma^T U^T$

$$\boxed{A^T} = \boxed{V} \boxed{\Sigma^T} \boxed{U^T}$$

(b) Given $A \in \mathbb{R}^{n \times n}$ invertible, $A = U \Sigma V^T$, can you find SVD of A^{-1} ?

$$A^{-1} = (V^T)^{-1} \Sigma^{-1} U^{-1} = V \Sigma^{-1} U^T$$

$$\Sigma^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & 1/\sigma_n & \\ & & 0 \end{bmatrix} \rightsquigarrow \text{diagonal elements are in the wrong order}$$

(c) Given $A \in \mathbb{R}^{n \times n}$, $A = U \Sigma V^T$, can you find SVD of A^2 ?

$$A^2 = A \cdot A = U \Sigma V^T U \Sigma V^T \neq U \Sigma^2 V^T \quad \text{spectral decomp}$$

(d) Given $A \in \mathbb{R}^{n \times n}$ symmetric, $A = U D U^T$, can you find an SVD of A ?

$$A = \begin{bmatrix} U \\ \end{bmatrix} \begin{bmatrix} |\lambda_1| & & \\ & \ddots & \\ & & |\lambda_n| \end{bmatrix} \begin{bmatrix} \text{sign}(\lambda_1) u_1^T \\ \vdots \\ \text{sign}(\lambda_n) u_n^T \end{bmatrix} \quad (\text{up to reordering the diagonal entries in non increasing order})$$

(e) If $A = x y^T$, $x, y \in \mathbb{R}^n$, what are the nonzero singular value and the corresponding singular vectors?

$$A = \begin{bmatrix} x \\ \vdots \\ x \end{bmatrix} \cdot \begin{bmatrix} |x||y| \\ \vdots \\ |x||y| \end{bmatrix} \cdot \begin{bmatrix} y^T / |y| \\ \vdots \\ y^T / |y| \end{bmatrix}$$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $u_1 \quad \quad \quad \sigma_1 \quad \quad \quad v_1^T$

→ thin / full SVD

Def: Given $A = U \Sigma V^T$ rank r , then the pseudoinverse is defined as $A^+ = V \Sigma^+ U^T$, $\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_r & & 0 \end{bmatrix}$ (*)

Remark: Last time: if A has full column rank we defined $A^+ = (A^T A)^{-1} A^T$

$$A = \begin{bmatrix} \text{matrix} \\ \end{bmatrix} = \begin{bmatrix} U \\ \end{bmatrix} \begin{bmatrix} \Sigma \\ \end{bmatrix} \begin{bmatrix} V^T \\ \end{bmatrix} \quad (*) \rightarrow A^+ = V \Sigma^+ U^T = V \Sigma^{-1} U^T$$

$\downarrow$
 invertible if A has full col. rank
 has orthonormal columns

$$(A^T A)^{-1} A^T = (V \Sigma \underbrace{U^T U}_I \Sigma V^T)^{-1} V \Sigma U^T = (V \Sigma^2 V^T)^{-1} V \Sigma U^T$$

$$= V \Sigma^{-2} \underbrace{V^T V}_I \Sigma U^T = V \Sigma^{-1} U^T \rightarrow \text{the same thing that we get with the new definition!}$$

Matrix norms

Def: A function $\|\cdot\| : \{\text{matrices}\} \rightarrow \mathbb{R}$ is a matrix norm if it satisfies all the following properties:

- (a) $\|A\| \geq 0$ for all A , $\|A\| = 0 \Leftrightarrow A = 0$
- (b) $\|\alpha A\| = |\alpha| \cdot \|A\| \quad \forall \alpha \text{ scalar}, \forall A$
- (c) $\|A + B\| \leq \|A\| + \|B\|$ for all A, B (triangle inequality)
- (d) $\|AB\| \leq \|A\| \cdot \|B\|$ for all A, B (sub-multiplicativity)

Def: (induced matrix norm): Given a vector norm $\|\cdot\|$ we can define

$$\|A\| = \max_{v \in \mathbb{R}^n, \|v\|=1} \frac{\|Av\|}{\|v\|}$$

→ if $\|\cdot\|$ an induced matrix norm, then for any A , for any v

$$\text{we have } \|Av\| \leq \|A\| \cdot \|v\|$$

the corresponding induced matrix norm

a chosen
vector norm

* The norm induced by $\|\cdot\|_2$ (Euclidean norm) is called 2-norm or spectral norm.

Additional property of the spectral norm: if Q_1, Q_2 are $n \times n$ orthogonal and $A \in \mathbb{R}^{n \times n}$ then $\|Q_1 A Q_2\|_2 = \|A\|_2$

$$\begin{aligned} \hookrightarrow \text{why? } \|Q_1 A Q_2\|_2 &= \max_{v \neq 0} \frac{\|Q_1 \overbrace{A Q_2 v}^z\|_2}{\|v\|_2} = \max_{v \neq 0} \frac{\|\overbrace{A Q_2 v}^z\|_2}{\|Q_2 v\|_2} \\ &= \max_{z \neq 0} \frac{\|A z\|_2}{\|z\|_2} = \|A\|_2 \end{aligned}$$

SVD of A

Consequence: $\|A\|_2 = \|U \Sigma V^T\|_2 = \|\Sigma\|_2 = \sigma_1$.

* Frobenius norm $A = \begin{bmatrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \end{bmatrix}$ $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$

Also the Frobenius norm is "unitarily invariant":

if Q_1, Q_2 are orthogonal then $\|Q_1 A Q_2\|_F = \|A\|_F$

$$\Rightarrow \|A\|_F = \|U \Sigma V^T\|_F = \|\Sigma\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}$$